

Vegetation restoration governs the changes in soil preferential flow and matrix infiltration

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ABSTRACT

Soil infiltration consists of preferential flow and matrix flow, both of which play a critical role in regulating soil water redistribution and the hydrological cycle. However, how different vegetation restoration types influence the partitioning between preferential flow and matrix infiltration remains poorly understood. Three typical vegetation restoration types of Moso bamboo pure forest, Moso bamboo-Chinese fir mixed forest and Chinese fir pure forest in the subtropical regions of southern China were selected, and the soil preferential flow and matrix flow were measured by using the improved surface-mounted double-ring infiltrometer. The effects of the driving factors on the preferential flow and matrix infiltration were quantified.

Results: 1) The preferential flow were 2.31–4.36 times greater than the matrix infiltration, accounting for 79.5%–81.3 % of the total infiltration (TIA) in Moso bamboo pure forest, 74.9 %–77.0 % of the TIA in Moso bamboo–Chinese fir mixed forest, and 69.8 %–72.5 % of the TIA in Chinese fir pure forest; 2) The total infiltration (560.54–739.47 mm) and preferential flow (425.68–595.81 mm) followed the order: Moso bamboo pure forest > Moso bamboo–Chinese fir mixed forest > Chinese fir pure forest. The cumulative matrix infiltration (136.62–184.20 mm) followed the order: Chinese fir pure forest > Moso bamboo–Chinese fir mixed forest > Moso bamboo pure forest. Fine root biomass (<2 mm), NCP, and BD were the dominant factors influencing the preferential flow, jointly accounting for 55.3 % of the contribution. Fine root biomass (<2 mm), clay, and BD showed close correlations with matrix infiltration, collectively explaining 60.6 % of the contribution. The findings provide mechanistic insights into soil hydrological functioning under different restoration strategies and offer practical implications for optimizing vegetation management and improving water conservation in subtropical ecosystems.

1. Introduction

Soil infiltration is an indispensable component of the forest hydrological cycle, exerting a substantial influence on pivotal ecohydrological processes, including surface runoff, soil holding, and solute transport. Vegetation restoration has been demonstrated to alter the composition of the below-ground root system and the soil properties of forest ecosystems (Zhao et al., 2018; Singh et al., 2013). These changes in turn affect soil infiltration patterns, water redistribution, and soil hydrological cycle processes. Consequently, an exhaustive examination of soil infiltration characteristics and their modulating mechanisms under diverse vegetation restoration types furnishes a substantial theoretical foundation for enhancing regional water resource management and sustainable forestry development.

Soil infiltration occurs primarily through two distinct ways:

preferential flow and matrix flow (J. Zhang et al., 2019). Preferential flow refers to the rapid and non-uniform movement of water through soil macropores (Fan et al., 2020). This process significantly influences surface runoff dynamics and serves as a critical part for groundwater recharge, playing a vital role in the hydrological cycle and ecosystem water balance. In contrast, matrix flow involves the slower movement of water through the soil matrix, governed by capillary forces. The water infiltrating via matrix flow contributes to soil moisture retention and serves as the primary source of plant-available water for root uptake. As a key component of the vegetation-soil-infiltration interaction, plant root systems modulate soil infiltration through multiple pathways. On the one hand, plant roots affect water infiltration by changing various soil properties such as soil bulk density, water content, and porosity (Zhu et al., 2020). On the other hand, the root system induces changes in soil properties, including soil organic carbon, aggregates,

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microorganisms, through growth and secretion, which can promote or inhibit water infiltration (Patra et al., 2022). It has been shown that changes in root length and root diameter metrics due to root growth and decay can lead to an increase in connected soil macropores (Cai et al., 2024), which can be a critical factor influencing preferential flow. However, when root growth and distribution is excessively dense, it can squeeze the original soil structure causing pore blockage, reducing water flow channels in the soil and hindering infiltration. As demonstrated by related studies, the capacity for infiltration is superior in soils from woodlands than in those from agricultural land (Hu et al., 2024), and is promoted more effectively by trees and shrubs than by herbs (Chandler et al., 2018). The infiltration rates of herbs with taproots exceed those of fibrous roots (Wu et al., 2017). The current study mainly focuses on the effect of vegetation restoration type on soil infiltration rather than its proportion. And it is evident that scholars in the relevant field have

directed their attention towards the impact of vegetation restoration on soil preferential flow, only focused on analysis of the distribution of preferential flow paths or the staining area. However, there is a paucity of quantitative studies that examine the infiltration process and its magnitude of soil matrix flow and preferential flow under different types of vegetation restoration. This limitation greatly restricts our understanding of soil infiltration patterns under various forms of vegetation restoration.

Soil infiltration is a variable and intricate process, exacerbated by the rapid non-uniform nature of soil infiltration patterns and the high variability of root-pore binary structure. How to accurately measure the changing characteristics of soil infiltration patterns is one of the important problems facing the field of soil hydrology. At present, researchers mainly measure matrix flow by using disc infiltrometer, constant head method or variable head method (Smettem and Kirkby, 1990;

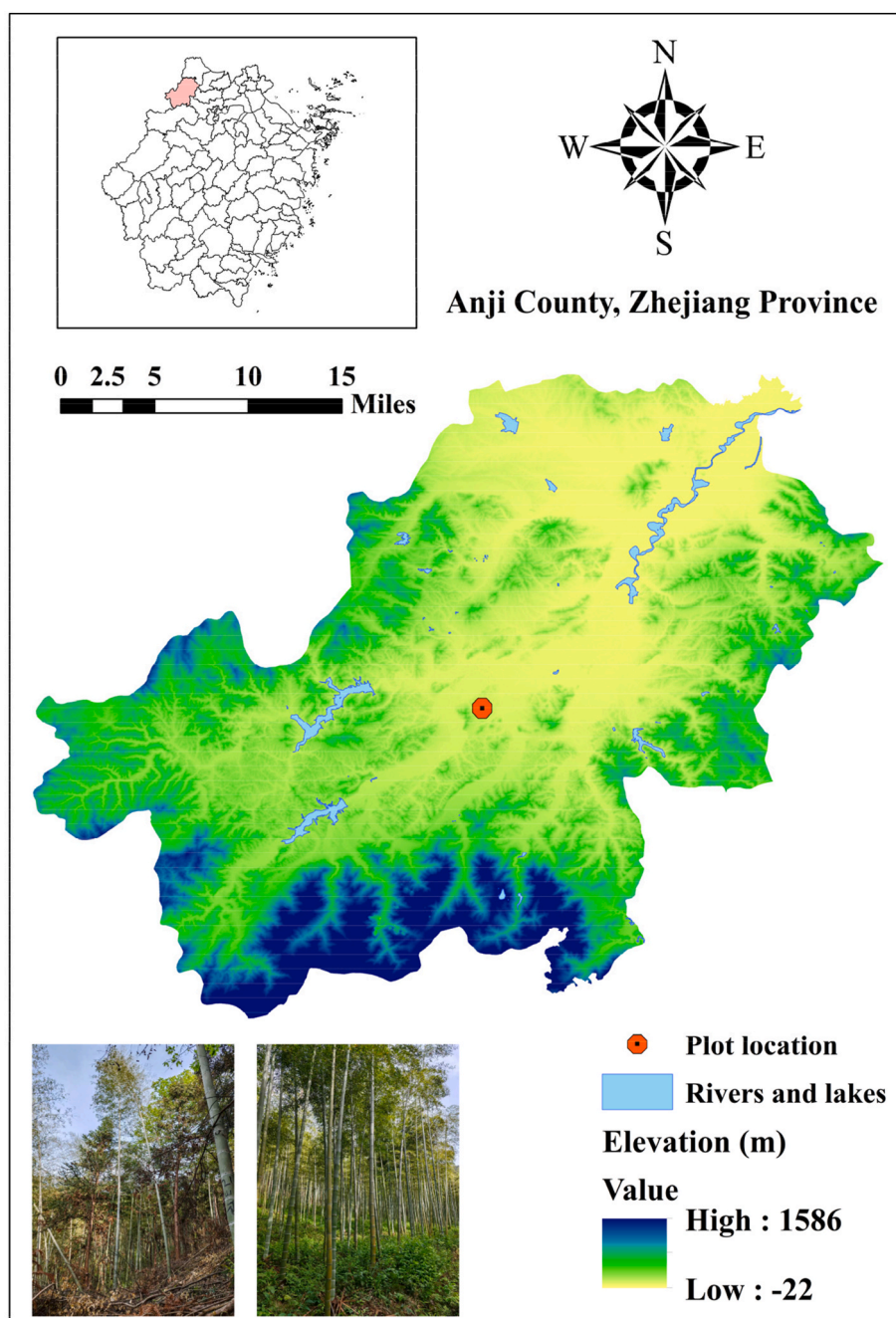


Fig. 1. Field experiment sample plots.

Zhang et al., 2006), which may damage the soil structure and affect the accuracy of measurement results. Dye tracers and ground-penetrating radar have applied to measure preferential flow (Zhao et al., 2023), which can only determine its the path and distribution, but cannot accurately determine the process and accumulative infiltration of soil preferential flow. Therefore, there is an urgent need for a method that can measure the process and related indicators of preferential flow and matrix flow in soil. As the research further develops, a way has been proposed to quantify soil matrix flow and preferential flow by surface-mounted ring infiltrometer combined with staining tracer technique (Zhang et al., 2017), and used to quantify the process of preferential flow and matrix flow in acacia forests on the Loess Plateau. Moreover, the process and magnitude of soil preferential flow and matrix flow were determined for eucalyptus forests based on a modified table-mounted ring infiltrometer (Zhao et al., 2022). Accordingly, it is highly desirable to measure preferential flow and matrix flow using a benchtop ring infiltrometer.

In this study, the three typical vegetation restoration types (Moso-bamboo pure forest, Moso-bamboo-Chinese fir mixed forest and Chinese fir pure forest) in the subtropics of southern China were selected. The characteristics of soil infiltration, preferential flow and matrix infiltration for the different vegetation restoration types were determined, and the main objectives of this study were as followed: (1) to quantitatively assess the characteristics of total soil infiltration, preferential flow and matrix flow under the different vegetation restoration types; (2) to elucidate the effects of root system characteristics and soil properties on soil infiltration, preferential flow and matrix flow.

2. Materials and methods

2.1. Study area

This study was conducted in the National Soil and Water Conservation Demonstration Park (30°36′30.3″N, 119°35′1.455″E) located in Anji County, Huzhou City, Zhejiang Province, as shown in Fig. 1. The site was selected due to its extensive and diverse natural forests, making it an ideal research area. Known as the “Bamboo County of China,” Anji is a core region for the country's bamboo industry, with approximately 600 km² of bamboo forests and abundant Moso bamboo (*Phyllostachys edulis*) resources. The region experiences a northern subtropical monsoon climate, characterized by a mean annual temperature of 15.5 °C and abundant precipitation (1485 mm annually), primarily concentrated in summer. Forest coverage exceeds 70 %, with dominant vegetation types including Moso bamboo forests and evergreen broad-leaved forests. The topography is predominantly mountainous and hilly, with higher elevations in the southwest and lower elevations in the northeast, forming a basin-like terrain surrounded by mountains on three sides and open to the northeast. The Tianmu Mountain Range extends into the southwestern part of the county, flanking its borders. The soils are primarily red and yellow earths derived from weathered sedimentary and acidic igneous rocks. Soil thickness reaches approximately 1 m in valleys and gentle slopes, while steep slopes exhibit thinner layers of 40–60 cm.

2.2. Selection and setting of research sample sites

In this study, three regionally representative natural forest types were selected as research objects: (1) Moso bamboo pure forest (*Phyllostachys edulis*), (2) Chinese fir pure forest (*Cunninghamia lanceolata*), and (3) 1:1 Moso bamboo-Chinese fir mixed forest. Among these, the Moso bamboo pure forest and the mixed forest had undergone succession within the past five years, while the Chinese fir pure forest was 20 years old. Field data were collected using handheld GPS devices and geological compasses. All experimental plots were located within the same demonstration park and were selected to ensure similar geomorphological conditions, including slope gradient, aspect, elevation, and

parent material, thereby minimizing large-scale pedological variability. According to local soil survey data, the soils in the study area are classified as yellow soils developed from comparable parent materials. Nevertheless, we acknowledge that small-scale spatial heterogeneity in soil particle-size distribution exists under natural field conditions. Detailed site selection criteria are provided in Table 1.

2.3. Soil infiltration experiments in the field

Three days after the end of the previous rainfall event, the soil infiltration experiment was carried out and completed within two days. For each of the three representative vegetation restoration types, four replicate experimental plots (3 m × 1 m) were prepared in root-free zones by removing surface litter to expose bare mineral soil. A surface-mounted double-ring infiltrometer system was employed to quantify preferential and matrix flows. The experiment was divided into two groups, one group added 4 g/L of brilliant blue dye solution to the inner ring while adding water to the outer ring to determine total soil infiltration. The other group evenly placed 3 cm of fine sand in the inner and outer rings to shield the preferential flow, then also added 4 g/L of brilliant blue dye solution to inner and water to outer ring to determine soil matrix flow, as shown in Fig. 2. The principle of blocking soil macropores by fine sand has been widely demonstrated (Zhao et al., 2022; Zhang et al., 2017). In terms of instrument arrangement, the double-ring infiltrometer was inserted into the soil 3 cm below the surface, which was consisted of an inner ring (D = 30 cm, H = 10 cm), and an outer ring (D = 60 cm, H = 10 cm). The gaps on both sides of the inner and outer rings close to the soil were filled and compacted with adjacent soil to prevent lateral leakage of water. Nylon mesh and foam boards were then laid on the surface of the soil to provide the buffer. At the beginning of the experiment, water was injected into the area encompassed by the inner and outer rings in a synchronized and rapid manner until it attained the 5 cm head marking position. Throughout the experiment, the head height of the inner and outer rings was maintained at a constant level. Timing started when water filling began, and the time taken was recorded every time the head height of the liquid level drops by 0.5 cm. When the water level dropped to 1 cm head height, the water level was replenished to 5 cm. Continuous recordings were made until the liquid level dropped continuously by 2 cm and the time between each 0.5 cm interval was almost the same, then the soil infiltration was considered to have reached the stable infiltration stage. The experiment was uniformly extended to 90 min after stabilization.

The initial infiltration rate (IIR), stable infiltration rate (SIR), and total infiltration amount at 90 min (TIA) were calculated for soil infiltration and matrix flow, respectively, based on the data recorded in the two sets of experiments. Furthermore, preferential flow indicators were derived based on the composition of soil infiltration proportion. The formulae for the calculations are given below:

$$f_0 = \frac{Q_0}{A \cdot t_0} \quad (1)$$

$$f_c = \frac{Q_c}{A \cdot t_c} \quad (2)$$

$$P = \frac{Q_0 + Q_c}{\pi \cdot r^2} \quad (3)$$

$$Q(t) = Q_m(t) + Q_p(t) \quad (4)$$

Where f_0 is the initial infiltration rate (mm/h); Q_0 is the amount of water infiltrated during the initial time period (mm³); A is the infiltration cross sectional area (mm²); t_0 is the initial time period (h); f_c is the stable infiltration rate (mm/h); Q_c is the amount of water infiltrated during the steady state (mm³); t_c is the steady state time period; P is the total infiltration amount (mm); r is the inner ring radius (mm); $Q(t)$ is all the water that enters the soil; $Q_m(t)$ is the water that infiltrates into the soil

Table 1

Basic information of the experiment area.

Vegetation restoration type	Altitude (m)	Slope	Aspect	Stand density	Mean tree height (m)	Mean diameter at breast high (cm)	Crown density
Moso bamboo	73	8°	West	2100	16	12	0.75
Mixed forest	75	10°	Southwest	1350	12	10	0.85
Chinese fir	76	6°	Southwest	925	10	10	0.6

**Soil infiltration**

Adding dye solution to inner ring and water to outer ring

**Matrix flow**

Placing fine sand on the surface soil, then adding dye solution to inner ring and water to outer ring

Fig. 2. Methods for measuring soil infiltration and matrix flow with the improved surface-mounted double-ring infiltrometer.

through the soil matrix; $Q_p(t)$ is the water that enters the soil through the preferential channel.

2.4. Collection and analysis of soil samples for experimentation

Twenty-four hours after the completion of the double ring infiltration experiment, the experiment setup was removed and the inner ring profile ($S = 30 \text{ cm} \times 50 \text{ cm}$) was excavated in the direction of the slope contour. Subsequently, soil samples from 0 to 50 cm were collected along the profile using a ring cutter with a diameter of 6 cm and a height of 10 cm. The profile was stratified vertically at 10 cm intervals to collect soil samples from 0 to 50 cm, with three soil samples per stratum. After sampling was completed, the samples were returned to the laboratory for the determination of basic physicochemical properties, including initial soil water content, saturated water content, soil bulk weight, total soil porosity, and soil non-capillary porosity.

The ring knife method was used to determine soil capacity and porosity. Undisturbed soil from the sample site was collected using a ring knife, which was inserted into the soil until it was completely filled. The collected soil samples, together with the ring knives, were placed in trays filled with water and left to saturate, and were carefully removed and weighed after 24 h. The soil samples were left to stand for 2 h to weigh and record. Then they were dried in an oven set at 110°C for 24 h until the soil was completely dry, and weighed again. Finally, soil bulk density (BD), total soil porosity (TP), soil capillary porosity (CP) and non-capillary porosity (NCP) were calculated by the formula.

A portion of the sample was sieved (2 mm) and the soil particle composition (sandy, loamy, clay) was measured using a laser particle size analyzer (MS 3000). The soil suspension was prepared and the soil pH was determined using the pH meter method. Another part of air-dried samples was sieved (0.25 mm) and the carbon in soil organic matter was oxidized under acidic conditions using the potassium dichromate oxidation-external heating method (Walkley-Black method) to calculate the soil organic matter content (SOC). The soil total nitrogen content (TN) was calculated using the Kjeldahl nitrogen determination method (Kjeldahl method) by sulphury acid dissolution of organic and

inorganic nitrogen in the soil.

2.4.1. Root biomass

After excavating the inner ring profile in the direction of the slope contour, the root system was collected simultaneously from all layers under the inner ring extent. Roots were thoroughly washed with water to remove soil particles and retain the intact roots. Root diameter was measured using vernier calipers and classified according to radius ($<1 \text{ mm}$, $1\text{--}2 \text{ mm}$, $>2 \text{ mm}$). Afterwards, the air-dried roots were placed in an oven and dried at 70°C for 24 h until constant weight, and then removed and weighed using a precision balance to determine the dry weight of roots in different diameter classes. Finally, the root biomass per unit volume was calculated based on the weighing results.

$$\text{Root Biomass} = \frac{M_{\text{root}}}{V_{\text{ir}}} \quad (5)$$

Where M_{root} is the dry weight (g) of roots for each size class in certain layer of soil; V_{ir} is the volume (cm^3) of a layer of soil in the inner ring.

2.5. Statistical analysis

In this study, two-way analysis of variance (ANOVA) was employed to evaluate the impact of soil physicochemical properties and rooting characteristics at varying depths under different vegetation restoration types. Additionally, the Duncan's multiple polarity method was utilized to assess the variability of physicochemical properties and rooting characteristics within the same soil layer under diverse vegetation restoration types. Pearson correlation analysis was implemented to ascertain the interrelationships between soil physicochemical properties, rooting characteristics and soil infiltration capacity. The relative contributions of soil and root variables to changes in soil infiltration capacity were determined by means of redundancy analysis (RDA) and variance partitioning (VPA). These methods were conducted in SPSS Statistics 27.0.1 and R Studio.

3. Results

3.1. Differences in soil infiltration capacity among different vegetation restoration types

3.1.1. Soil total infiltration and its differences

The IIR and SIR of each vegetation restoration type followed the order: Moso bamboo pure forest > Moso bamboo-fir mixed forest > Chinese fir pure forest, and the Moso bamboo pure forest had the largest initial infiltration rate (950.61 mm/h) and the stable infiltration rate (364.59 mm/h). The IIR and SIR of Moso bamboo pure forest were found to be 1.08, 1.18 times higher than the Moso bamboo-Chinese fir mixed forest, and 1.09, 1.33 times higher than the Chinese fir pure forest, respectively, as shown in Fig. 3. During both the initial infiltration stage and the stable infiltration stage, significant discrepancies emerged between the IIR and SIR of the three vegetation restoration types. Notably, the SIR of the Moso bamboo pure forest exhibited an increase of 90.96 mm/h compared with that of the Chinese fir pure forest, which is a substantial difference ($P < 0.01$). Concurrently, the order of the three types was as follows: Moso bamboo pure forest > Moso bamboo-Chinese fir mixed forest > Chinese fir pure forest. The Moso bamboo pure forest showed the highest TIA (719.03 mm), exceeding the mixed forest and Chinese fir pure forest by 39.50 mm and 139.79 mm, respectively.

3.1.2. Preferential flow and its proportion

In soil infiltration patterns, preferential flow dominated soil infiltration, comprising 72.9 %–77.8 % (IIR), 59.7 %–75.7 % (SIR), and 71.4 %–80.6 % (TIA) of soil infiltration. The pure forest of Moso bamboo exhibited significant preferential flow characteristics, with its IIR, SIR,

and TIA accounting for the largest proportion. These proportions could be improved by 3.1 %–5.9 %, 6.3 %–15.9 %, and 4.8 %–9.2 %, respectively. The IIR, SIR and TIA of the preferential flow of each vegetation restoration type present a congruent proportion with that of soil infiltration: Moso bamboo pure forest > Moso bamboo-Chinese fir mixed forest > Chinese fir pure forest. The above infiltration indicators values of Moso bamboo pure forest were the largest, which were 733.58 mm/h, 275.71 mm/h, and 589.96 mm, respectively, and were markedly different from those of other vegetation restoration types ($P < 0.01$). The most significant disparity in SIR was observed between Moso bamboo pure forests and Chinese fir pure forests, which attained 40.7 %.

3.1.3. Matrix flow and its proportion

Matrix flow contributed a relatively small proportion to total soil infiltration across all vegetation restoration types, accounting for approximately 20–40 % of infiltration indices (IIR, SIR and TIA). Notably, the contribution of matrix flow consistently followed the order: Chinese fir pure forest > Moso bamboo-Chinese fir mixed forest > Moso bamboo pure forest, which was opposite to the pattern observed for preferential flow. Among the three vegetation types, Chinese fir pure forest exhibited the strongest matrix infiltration capacity, characterized by higher initial and steady infiltration rates and greater cumulative matrix infiltration. Compared with the other vegetation types, matrix flow in the Chinese fir pure forest increased substantially, with steady infiltration rate and cumulative infiltration enhanced by up to ~21 mm/h and ~34 mm, respectively. Differences between Chinese fir pure forest and Moso bamboo pure forest were statistically significant ($P < 0.01$), whereas matrix flow parameters in the mixed forest were comparable to those of the Chinese fir pure forest, with differences generally below

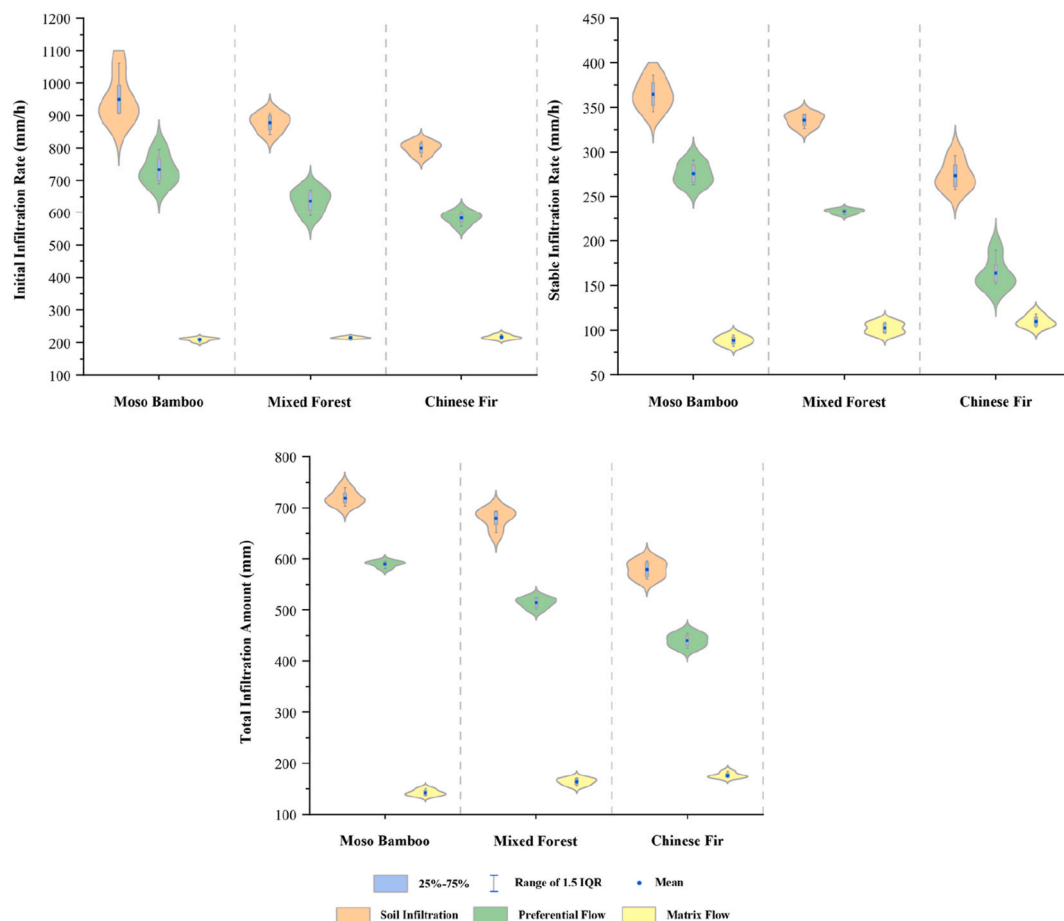


Fig. 3. Characteristics of soil total infiltration, preferential flow and matrix flow across different vegetation restoration types.

10 %.

In summary, the Moso bamboo pure forest demonstrated optimal performance in IIR, SIR and TIA of soil infiltration and preferential flow. Compared to the mixed and Chinese fir pure forests, it increased the initial infiltration rate by 13.5 % and 20.5 %, the stable infiltration rate by 20.9 % and 43.5 %, and the total infiltration amount by 15.0 % and 24.4 %, sequentially. In the matrix flow, when compared with the other two, the Chinese fir pure forest demonstrated superior performance, with enhancements of 1.9 %, 15.5 %, and 15.7 % observed in the IIR, SIR, and TIA, respectively.

3.2. Soil properties and root characteristics of different vegetation restoration types

Vegetation restoration markedly altered soil physical properties and root biomass distribution across the 0–50 cm soil profile (Fig. 4, Table 2). Along the restoration gradient from Chinese fir pure forest to mixed forest and Moso bamboo pure forest, soil bulk density

significantly decreased, particularly in the upper 0–30 cm layer, indicating improved soil structure under bamboo-dominated stands. The lowest bulk density (1.10 g cm^{-3}) was observed in the 0–10 cm layer of Moso bamboo pure forest. Non-capillary porosity showed strong depth- and vegetation variability. In the surface soil (0–10 cm), Moso bamboo pure forest exhibited significantly higher non-capillary porosity than Chinese fir pure forest, suggesting enhanced macropore development. In deeper layers (10–50 cm), non-capillary porosity followed the order: mixed forest > Chinese fir pure forest > Moso bamboo pure forest, with the mixed forest showing the greatest enhancement, particularly at 30–40 cm depth. Soil texture also varied among vegetation types, with higher clay contents in the mixed forest and Chinese fir pure forest compared with Moso bamboo pure forest, implying stronger water-holding capacity in non-bamboo stands.

Root biomass differed significantly among restoration types and soil depths. Moso bamboo pure forest and mixed forest exhibited substantially higher fine-root biomass (<2 mm) in the 0–20 cm layer than Chinese fir pure forest, indicating intensified root–soil interactions near

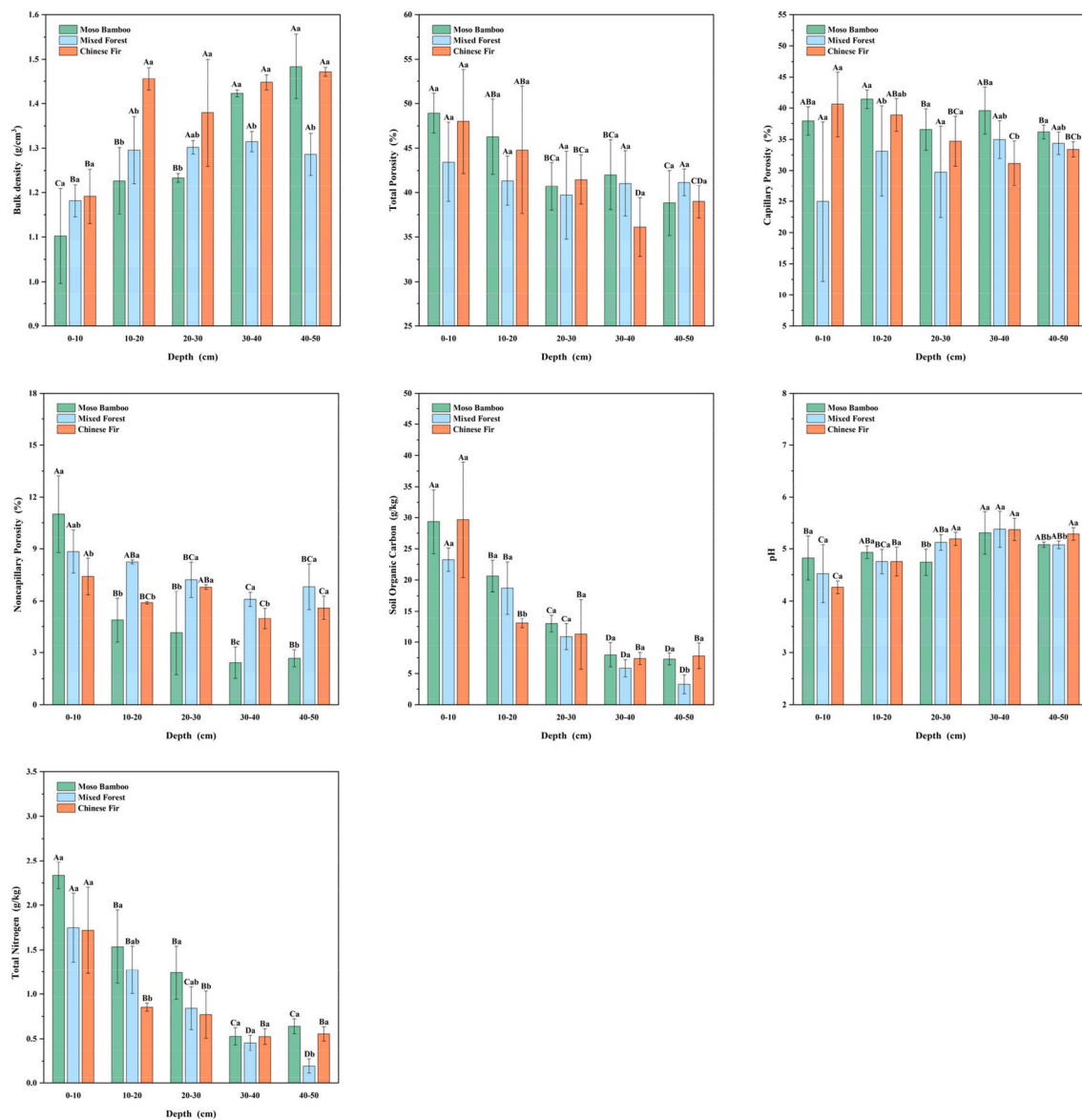


Fig. 4. Bulk density, total porosity, capillary porosity, noncapillary porosity, total organic carbon, pH, total nitrogen of 0–10, 10–20, 20–30, 30–40, 40–50 cm in Moso bamboo pure forest, Moso bamboo-Chinese fir mixed forest, Chinese fir pure forest. The different capital and lowercase letters above the bars indicate significant differences among the different soil depths of the same vegetation restoration type and the different vegetation restoration types at the same soil depth, respectively ($P < 0.05$).

Table 2

Soil particle composition and root biomass characteristics of different vegetation restoration types.

Soil Depth	Vegetation restoration type	sand (%)	silt (%)	clay (%)	RB (<1 mm) (g/m ³)	RB (1–2 mm) (g/m ³)	RB (>2 mm) (g/m ³)
0–10 cm	Moso bamboo	70.83 ± 2.68Aa	18.96 ± 2.58Da	10.21 ± 0.94Bb	6814.13 ± 2188.55Aa	1312.5 ± 268.18Aa	5561.76 ± 3847.46Aa
	Mixed forest	63.44 ± 7.22Aa	21.7 ± 7.91Ca	14.87 ± 2.1Aa	5060.86 ± 2692.07Aa	1056.17 ± 190.46Aa	6922.09 ± 9403.27Aa
	Chinese fir	69.68 ± 3.47Aa	20.61 ± 7.91Ca	9.71 ± 0.98Bb	2201.64 ± 60.4Ab	440.33 ± 205.37Ab	1408.87 ± 47.11Ab
10–20 cm	Moso bamboo	52.38 ± 10.89Bb	31.4 ± 12.68Ca	16.22 ± 2.13Aa	2977.26 ± 800.66Ba	786.05 ± 456.89Ba	7321.66 ± 6387.14Aa
	Mixed forest	70.3 ± 3.26Aa	17.19 ± 2.28Cb	12.51 ± 4.08Aa	2144.17 ± 642.71Ba	814.01 ± 27.07Ba	14692.37 ± 7384.86Aa
	Chinese fir	63.12 ± 3.86Bab	24.11 ± 2.28Cab	12.78 ± 2.48Aa	749.27 ± 373.28Bb	309.87 ± 164.34Ab	470.08 ± 306.11BCb
20–30 cm	Moso bamboo	48.23 ± 10.55Bb	45.74 ± 7.82Ba	6.03 ± 3.82BCb	893.04 ± 181.21Cab	211.32 ± 60.74Cb	262.08 ± 121.08Bb
	Mixed forest	63.77 ± 8.24Aa	22.49 ± 10.8Cb	13.74 ± 2.7Aa	1396.41 ± 284.11Ba	360.73 ± 111.07Ca	2387.59 ± 1744.67Ba
	Chinese fir	60.82 ± 6.45Bab	25.12 ± 10.8Cb	14.07 ± 1.03Aa	680.83 ± 302.01Bb	76.39 ± 28.67Bc	1096.4 ± 1014.75ABab
30–40 cm	Moso bamboo	32.21 ± 8.12Cb	62.35 ± 2.77Aa	5.44 ± 5.76BCb	455.54 ± 200.84Cb	198.41 ± 63.9Ca	171.18 ± 82.05Ba
	Mixed forest	43.49 ± 7Ba	40.92 ± 6.86Bb	15.59 ± 1.54Aa	1101.09 ± 173.77Ba	98.59 ± 24.54Db	209.55 ± 35.06Ba
	Chinese fir	49.93 ± 1.33Ca	35.57 ± 6.86Bb	14.51 ± 2.53Aa	596.65 ± 384.16Bb	28.29 ± 11.18Bc	0 ± 0Cb
40–50 cm	Moso bamboo	35.4 ± 5.62Ca	62.66 ± 5.66Ab	1.94 ± 0.98Cb	417.34 ± 213.46Cb	266.67 ± 69.63Ca	175.42 ± 104.52Ba
	Mixed forest	13.07 ± 1.31Cb	71.96 ± 1.88Aa	14.97 ± 0.7Aa	1078.72 ± 301.14Ba	82.64 ± 3.94Db	158.67 ± 118.61Ba
	Chinese fir	17.28 ± 5.12Db	67.61 ± 1.88Aab	15.12 ± 1.35Aa	324.76 ± 451.22Bb	23.48 ± 13.11Bb	0 ± 0Cb

Note: capital letters indicate that there are significant differences among different soil layers of the same vegetation restoration type ($P < 0.05$). At each soil layer, different lowercase letters within one column indicate a significant difference among the different vegetation restoration types ($P < 0.05$).

the surface. In contrast, the mixed forest showed greater fine-root biomass in deeper soil layers (30–50 cm), reflecting deeper root penetration and a more vertically distributed root system. Overall, vegetation restoration modified soil structure primarily through changes in root biomass distribution and porosity, thereby creating distinct subsurface environments for water infiltration.

3.3. Influencing factors of soil infiltration patterns of different vegetation restoration types

3.3.1. Soil total infiltration factors

Fine plant roots can improve soil physicochemical properties, thus enhancing soil infiltration capacity. RB (<1 mm) and RB (1–2 mm) significantly promoted the SIR and TIA of soil infiltration ($P < 0.01$). IIR, SIR and TIA showed significant correlation with BD and NCP ($P < 0.05$), in which NCP showed a facilitating effect, and BD showed a significant negative correlation, as illustrated in Table 3.

Variance partitioning analysis (VPA) was used to quantify the relative contribution of soil physicochemical properties and root characteristics to the regulation of soil total infiltration. In this study, all indicators were divided into two groups of soil physicochemical properties and root biomass for analysis, and the results are shown in Fig. 5. These two groups of variables can jointly explain 87.8 % of the variation in soil infiltration, and the unexplained portion is 12.2 %. In particular, soil physical and chemical properties and root characteristics jointly explained 22.9 % of the observed changes and independently explained 44.6 % and 20.3 % of the total changes, respectively ($P < 0.05$).

Redundancy analysis (RDA) was applied to explore the factors influencing soil total infiltration for each vegetation type and the results were shown in Fig. 6. The results showed that RDA1 and RDA2 could jointly explain 98.3 % of the variance of the response variables in soil infiltration ($P < 0.05$). BD, NCP, sand, RB (<1 mm), and RB (1–2 mm) could jointly explain 74.1 % of the variance of RDA1, and had the highest explanatory power of the three response variables of soil

infiltration, namely IIR, SIR, and TIA. The effects of BD and fine root biomass (<2 mm) were significant and opposite on the three infiltration indicators, with independent contributions of up to 16.5 % and 27.4 %, respectively. From the distribution of sample points, it can be seen that the pure forest of Moso bamboo was significantly and positively affected by the variables of NCP, sand, RB (<1 mm), RB (1–2 mm) and had the strongest soil infiltration capacity, while the pure forest of Chinese fir was negatively affected by BD and clay.

3.3.2. Preferential flow factors

The results of the study demonstrated that soil physicochemical properties and root characteristics exerted divergent effects on the preferential and matrix flows, as illustrated in Table 4. In the preferential flow, the effects of BD, TN, silt and clay was found to be highly significant on IIR ($P < 0.01$). Both SIR and TIA exhibited a high degree of significance in their response to BD, silt and clay ($P < 0.01$). Additionally, a significant positive correlation was also observed for NCP ($P < 0.05$). The three root biomass indicators exhibited a positive effect on the preferential flow, with RB (<1 mm) and RB (1–2 mm) demonstrating remarkably boost on SIR and TIA ($P < 0.01$), respectively. Furthermore, RB (<1 mm) also exerted an influence on IIR ($P < 0.05$).

The two groups of variables could jointly explain 90.3 % of the variation, and the unexplained variation only accounted for 9.7 %, which also had a high explanatory power, as shown in Fig. 6. Soil physicochemical properties and root biomass could explain 79.8 % and 49.6 % of the infiltration capacity of the soil preferential flow, respectively ($P < 0.05$). Nearly 80 % of the overall contribution of root biomass was shared with soil physical and chemical properties, and if the covariate contribution was excluded, soil physicochemical properties and root biomass contributed 40.8 % and 10.6 %, respectively, with the former independently contributing nearly four times more than root biomass.

In the preferential flow, RDA1 and RDA2 collectively accounted for 98.2 % of the variance in the response variables, with explaining 92.5 %

Table 3

Correlation analysis of soil physicochemical properties, root characteristics and soil infiltration characteristics under different vegetation restoration types.

Infiltration properties	BD	TP	CP	NCP	SOC	pH	TN	sand	silt	clay	RB (<1 mm)	RB (1–2 mm)	RB (>2 mm)
Initial infiltration rate	−0.832**	0.236	0.086	0.626*	0.482	0.404	0.802**	0.551	0.620*	−0.532	0.570	0.506	0.227
Stable infiltration rate	−0.929**	−0.051	−0.207	0.694*	0.098	0.451	0.490	0.513	0.618*	−0.562	0.779**	0.838**	0.467
Total Infiltration amount	−0.845**	−0.002	−0.118	0.663*	0.123	0.545	0.556	0.639*	0.678*	−0.549	0.802**	0.813**	0.529

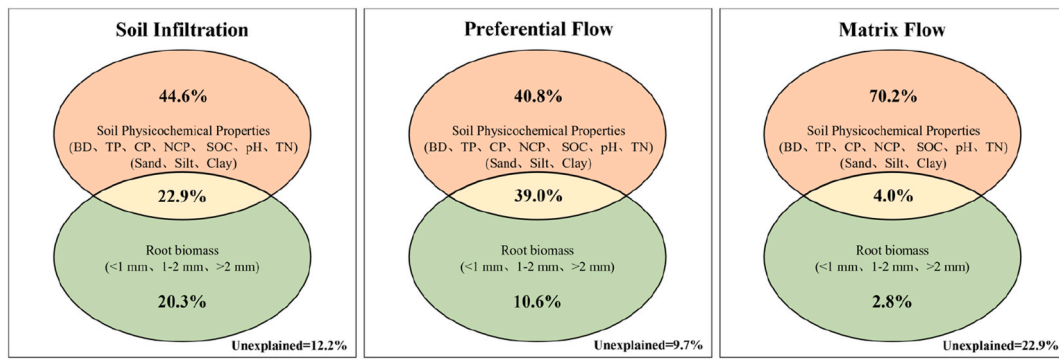


Fig. 5. Variance partitioning analysis of soil physicochemical properties and root characteristics on soil infiltration, preferential flow and matrix flow.

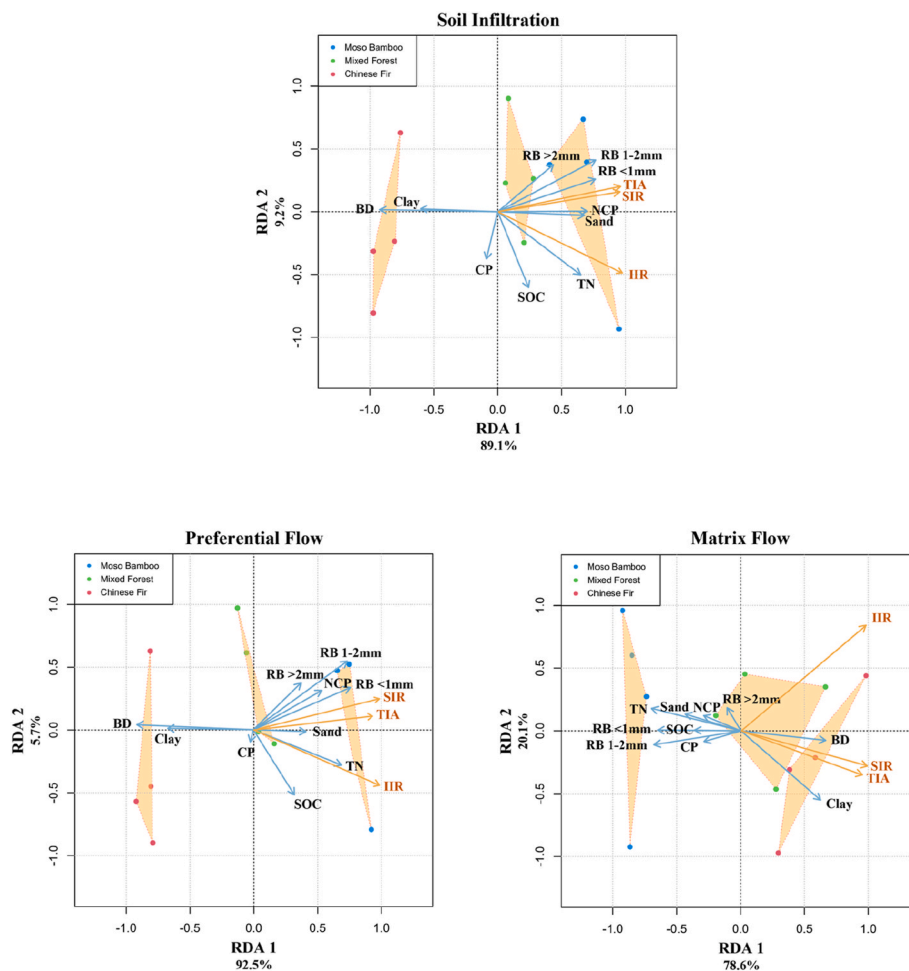


Fig. 6. Redundancy analysis (RDA) sequencing map based on soil physicochemical properties, root characteristics and soil infiltration characteristics. The physicochemical properties of soil and root characteristics are represented by blue vectors, while soil infiltration characteristics are denoted by brown vectors. The areas surrounded by blue, green and red dots represent the results to which different vegetation restoration types are affected by various indicators.

and 5.7 %, respectively. This finding demonstrated strong explanatory power ($P < 0.05$), as illustrated in Fig. 6. RDA1 primarily reflected the effects of BD, NCP, TN, sand, clay, RB (<1 mm), and RB (1–2 mm). The aforementioned variables collectively contributed to RDA1, accounting for 93.5 % of the variance of response variables, and 58.5 % of that in RDA2. The remaining explanatory variables contributed 41.5 % of its variance to RDA2. The cumulative contribution of the six variables of BD, NCP, sand, clay, RB (<1 mm), and RB (1–2 mm) to the infiltration capacity in both RDA1 and RDA2 amounted to 74.3 %. Notably, the independent contributions of BD, soil particle composition, and root

biomass were 16.6 %, 21.2 %, and 35.9 %, respectively. IIR in preferential flow was positively influenced by SOC and TN. In contrast, NCP, sand content, and root biomass showed significant positive correlations with SIR and TIA, while BD was negatively correlated with three soil infiltration indicators. The distribution of sample points in the redundancy analysis (RDA) ordination plot revealed distinct responses of the three vegetation restoration types to the explanatory variables. Moso bamboo pure forests exhibited the highest IIR, SIR, and TIA, primarily regulated by TN, RB (<1 mm) and RB (1–2 mm). In contrast, the mixed forest was predominantly influenced by RB (>2 mm), whereas BD and

Table 4
Correlation analysis of soil physicochemical properties, root characteristics and preferential flow, matrix flow characteristics under different vegetation restoration types.

Infiltration properties		BD	TP	CP	NCP	SOC	pH	TN	sand	silt	clay	RB (<1 mm)	RB (1–2 mm)	RB (>2 mm)
Preferential flow	Initial infiltration rate	–0.887**	0.063	–0.013	0.420	0.463	0.265	0.732**	0.001	0.845**	–0.800**	0.600*	0.515	0.227
	Stable infiltration rate	–0.910**	0.034	–0.121	0.645*	0.153	0.482	0.557	0.430	0.799**	–0.753**	0.785**	0.829**	0.434
	Total Infiltration amount	–0.827**	0.167	0.062	0.592*	0.308	0.537	0.693*	0.547	0.851**	–0.821**	0.797**	0.768**	0.410
Matrix flow	Initial infiltration rate	0.429	–0.484	–0.266	–0.130	–0.260	–0.347	–0.388	–0.580*	–0.491	0.662*	–0.466	–0.565	0.045
	Stable infiltration rate	0.577*	–0.410	–0.261	–0.257	–0.406	–0.466	–0.721**	–0.555	–0.752**	0.763**	–0.577*	–0.577*	–0.126
	Total Infiltration amount	0.688*	–0.373	–0.232	–0.346	–0.271	–0.572	–0.669*	–0.559	–0.854**	0.831**	–0.631*	–0.625*	–0.168

clay were the key factors governing preferential flow processes in Chinese fir pure forests.

3.3.3. Matrix flow factors

As shown in Tables 4 and IIR was mainly affected by sand and clay in matrix flow ($P < 0.05$). BD, TN, silt, and clay had significant effects on SIR and TIA ($P < 0.05$), with BD and clay positively regulating matrix flow to a highly significant level ($P < 0.01$), while TN and silt showed significant negative correlation. The three root biomass indicators also showed negative correlation on matrix flow, in which the reverse regulation of SIR and TIA by RB (<1 mm) and RB (1–2 mm) reached a significant level. ($P < 0.05$).

In the VPA analysis of matrix flow, the two groups of variables could jointly explain 77.1 % of the variation and 22.9 % of the unexplained, with strong explanatory power ($P < 0.01$), as shown in Fig. 5. If the covariate part is not considered, the soil physicochemical properties and the root biomass were able to explain 70.2 %, and 2.8 %, respectively, of the variation of matrix flow, which shows that the soil physicochemical properties are the dominant factors. In addition, the results also demonstrated that the total contribution of root biomass in the preferential flow was 7.29 times higher than that of the matrix flow, with an increase in the independent explanatory power of 7.9 %. While compared with preferential flow, the independent explanatory power of soil physicochemical properties in the matrix flow was significantly higher by 29.4 %.

RDA1 and RDA2 collectively accounted for 98.7 % of the variance in the response variable, demonstrating a high degree of explanatory power, with RDA1 and RDA2 exhibiting an explanatory power of 78.6 % and 20.1 %, respectively. This finding attained a statistical significance level ($P < 0.01$), as illustrated in Fig. 6. The combined effects of BD, TN, sand, clay, and RB (1–2 mm) accounted for 72.3 % of the observed variation. It was found that BD, soil particle composition and root biomass had a prominent contribution to the matrix flow, a cumulative total of 77.5 %. The analysis indicated a significant positive effect of BD and clay, with independent contributions of 13.5 % and 20.8 %, respectively. In the case of IIR, BD and RB (1–2 mm), a significant effect on the infiltration process was observed. By contrast, SIR and TIA were primarily regulated by the multiple variables of BD, TN, clay and RB (<1 mm). Furthermore, Chinese fir pure forest demonstrated optimal performance in matrix flow, where IIR, SIR and TIA were mainly dominated by BD, sand and clay.

4. Discussion

4.1. Characteristics of soil physicochemical properties and roots of different vegetation restoration types

Vegetation restoration is an effective way to improve soil quality and regulate soil hydrological processes in the humid subtropical regions of southern China (Hao et al., 2019; Zhang et al., 2019), its differences in types can lead to significant changes in soil physicochemical properties and root systems (Gu et al., 2019). Previous studies have shown that soil physical properties and root systems are significantly affected by vegetation restoration (Feng et al., 2021; Gu et al., 2018). Root growth creates root channels, resulting in increased soil porosity at the root-soil interface and loosening of the soil texture, which is ultimately reflected in improved soil capacity, soil porosity and root biomass (Helliwell et al., 2017). The presence of Moso bamboo leads to a decrease in shallow soil bulk weight and an increase in non-capillary pore space (Chen et al., 2024). The reasons for this are usually attributed to the distinctive root system configuration of Moso bamboo, its well-developed fine roots and the prominent role of its distribution in the above mentioned aspects (Kaushal et al., 2019). The present study has revealed significant variations in the soil's physical characteristics and root biomass among the three types of vegetation studied. In comparison with the Chinese fir pure forest, observations in the top 30 cm of

soil revealed notable changes in the BD, NCP, and root biomass of both the Moso bamboo pure forest and the mixed forest. The change observed in the Moso bamboo pure forest was particularly pronounced, which significantly reduced the soil bulk density from 10 to 30 cm and simultaneously increased the noncapillary porosity and root biomass from 0 to 30 cm. The latter two showed the opposite changes to the soil bulk density, reflecting the role of Moso bamboo root system. In the shallow soil layer where the root system was not distributed or less distributed, the bamboo whips loosened the soil through lateral extension and the attached dense fine roots, creating more root channels and causing changes in soil bulk density and soil porosity. Meanwhile, the present study further demonstrated that different tree species in bamboo-broad-leaved mixed forests can significantly increase fine root biomass (Zhou et al., 2022). The mixed forest share some of the common characteristics of the two pure forests. Although the root biomass of Moso bamboo decreased sharply with soil depth, the effect of the root system of Chinese fir increased, owing to the combined action of the two root systems, which not only reduced the soil bulk weight in the 10–30 cm layer, but also reduced the soil bulk weight in the deeper layer of the 30–50 cm layer. Compared with other vegetation restoration types, the non-capillary porosity and root biomass (<1 mm) also remained high in all five soil layers from 0 to 50 cm, and the fine root biomass was much higher in the 20–30 cm layer, which was in line with the characteristics of the root distribution of the mixed forests, and consistent with the results of the previous studies (Ma et al., 2022; Zhou et al., 2022).

It has been established through preceding research that alterations in soil chemical properties are the consequence of physical and chemical interactions (Delory et al., 2023). Changes in the soil environment caused by the accompanying expansion of the plant root systems and the secretion of organic compounds from the roots usually lead to changes in the chemical properties such as organic carbon and pH (Paterson, 2003; Read et al., 2003; Li et al., 2023). In the subtropics, compared to the broadleaf pure and mixed forests, Moso bamboo pure forest can increase soil pH while decreasing organic carbon, total nitrogen (Ma et al., 2022). In this paper, significant higher levels of soil organic carbon (SOC) and total nitrogen (TN) in the Moso bamboo pure forest were found in the 10–20 cm soil layer compared to the Chinese fir pure forest. However, no significant differences in SOC, pH and TN were observed in the other soil layers. The significant differences in the 10–20 cm soil layer can be attributed to the distribution of the Moso bamboo root system, which is characterized by its shallow distribution (Guo, 2011). The effects of the three vegetation restoration types on chemical properties were found to be convergent in the other soil depths.

In summary, vegetation restoration can lead to significant differences in soil physical properties and root characteristics (Zhang et al., 2024). Specifically, the Moso bamboo pure forest in the 0–30 cm shallow soil layer greatly reduced soil bulk density and increased non-capillary porosity, organic matter and total nitrogen. Although the mixed forest had the same effect on the above indexes, the degree of effect was lower than that of the Moso bamboo pure forest. Notably, the area of soil affected is wider, and the positive effect brought by its root characteristics can be extended to the deep soil, showing the spatial effect of soil improvement, and its spatial heterogeneity contrasts with that of the Moso bamboo pure forest that is a single superior species community.

4.2. Factors influencing preferential and matrix flows

The influence of soil physicochemical properties and root characteristics on soil infiltration has been widely documented (Zimmermann and Elsenbeer, 2008; Hao et al., 2019; Lozano-Baez et al., 2019; Wang et al., 2022); however, disentangling the dominant controls on preferential flow and matrix infiltration remains a key challenge, particularly under different vegetation restoration types. In the present study, preferential flow accounted for 71.4 %–80.6 % of total infiltration, confirming its dominant role in soil water transport in subtropical forest

ecosystems. This magnitude is consistent with recent field-based infiltration studies in forested hillslopes, which report preferential flow contributions exceeding 65 % under well-structured soils (Zhang et al., 2019, 2024; Li et al., 2023). Compared with matrix flow, preferential flow exhibited much higher infiltration rates and cumulative infiltration, highlighting its efficiency in transmitting water through macropore networks. Our results demonstrate that preferential flow was primarily controlled by fine root biomass (<2 mm), non-capillary porosity, and bulk density, whereas matrix infiltration was more strongly associated with fine root biomass, clay content, and bulk density. These contrasting controls indicate that preferential flow and matrix flow respond to distinct structural and textural attributes of the soil. Similar patterns have been reported in recent studies showing that preferential flow is highly sensitive to macropore abundance and continuity, while matrix flow is constrained by soil texture and micro-pore connectivity (Chen et al., 2024; Ju et al., 2024; Wang et al., 2024). Soil texture constitutes a fundamental constraint on soil hydraulic behavior and is generally stable over decadal timescales (Zhao et al., 2023). The higher clay content observed in the Chinese fir forest, particularly in surface soils, likely contributed to its lower saturated conductivity and greater matrix flow contribution. However, our results indicate that vegetation restoration regulates infiltration primarily through structural modification rather than changes in particle-size distribution. Preferential flow was strongly associated with non-capillary porosity, bulk density, and fine-root biomass, whereas clay content exerted a stronger control on matrix flow. This suggests that soil texture defines the baseline hydraulic framework, while root-induced pore formation, aggregation, and connectivity determine the dominance of preferential versus matrix flow pathways. Such a distinction is critical for understanding hydrological responses to vegetation restoration under natural soil heterogeneity. The stronger dependence of matrix flow on clay content observed here reflects the role of fine particles in regulating capillary pore continuity and hydraulic resistance, which limits infiltration under dense or fine-textured conditions.

Although root biomass was significantly correlated with preferential flow, living roots themselves do not directly function as flow conduits. Instead, preferential flow is more likely facilitated by root-induced macropores formed through root growth, shrinkage, and decay, as well as by gaps at the root–soil interface (Cui et al., 2019; Shi et al., 2021). Recent X-ray CT and dye-tracer studies have shown that macropores associated with decayed roots and rhizosphere detachment provide continuous preferential pathways that persist beyond the lifespan of living roots (Feng et al., 2020). In bamboo-dominated systems, the rapid turnover of fine roots and the dense rhizome network may accelerate the formation of such macropores, thereby enhancing preferential flow even when living root biomass fluctuates seasonally (Zhao et al., 2025). Furthermore, root activity indirectly modifies soil structure by reducing bulk density, increasing organic matter inputs, and promoting aggregate stability (Wu et al., 2016; Li et al., 2023). These processes enhance pore connectivity and enlarge pore sizes, which collectively favor preferential flow development. In contrast, matrix infiltration is less influenced by large structural pores and is instead governed by the distribution and connectivity of small pores within soil aggregates. This explains why Chinese fir pure forests, despite exhibiting lower preferential flow, showed relatively higher matrix infiltration compared with bamboo-dominated systems. Similar trade-offs between preferential and matrix flow have been observed in mixed and coniferous forests, where finer-textured soils and higher clay contents suppress macropore dominance and promote matrix-controlled infiltration (Sun et al., 2023).

Overall, our findings suggest that vegetation restoration alters soil infiltration not only by changing root biomass, more importantly, by regulating pore connectivity, and the persistence of root-derived macropores. Preferential flow is mainly driven by structurally induced macropores associated with root systems and non-capillary porosity, whereas matrix flow is constrained by soil texture and micro-pore

architecture. These mechanisms are broadly consistent with recent hydrological studies but further highlight the unique role of bamboo root systems in amplifying preferential flow pathways. Such differences have important implications for water redistribution, solute transport, and slope hydrological responses in subtropical forest landscapes.

4.3. Methodological limitations and implications

Although the surface-mounted double-ring infiltrometer combined with a sand-layer separation technique has been widely applied to distinguish preferential flow from matrix infiltration, several limitations should be acknowledged. The introduction of a 3 cm fine sand layer inevitably adds an artificial hydraulic resistance at the soil surface, which may influence infiltration dynamics, particularly during the initial infiltration phase. This resistance can lead to a slight underestimation of absolute matrix flow rates compared with undisturbed conditions. It should also be noted that the sand-layer method isolates matrix flow operationally rather than mechanistically. While it effectively suppresses macropore-dominated preferential pathways, it does not fully account for microscale root–soil interface flow or partially connected macropores that may still contribute to infiltration under natural conditions. Therefore, the reported matrix flow represents a conservative estimate of soil matrix infiltration. Despite these limitations, the method remains appropriate for comparative analyses across vegetation restoration types, which is the primary objective of this study. Future work combining dye tracing, X-ray CT imaging, or *in situ* soil moisture monitoring could further refine the quantification of preferential and matrix flow processes and reduce uncertainties associated with surface-based separation techniques.

5. Conclusions

This study systematically compared soil preferential flow and matrix infiltration under three typical vegetation restoration types in subtropical China and elucidated their controlling mechanisms from the perspective of soil properties and root characteristics. The results demonstrate that vegetation restoration fundamentally alters soil infiltration patterns by regulating pore structure, root distribution, and soil physical properties, leading to pronounced differences in the partitioning between preferential flow and matrix flow. Preferential flow dominated soil water infiltration across all vegetation types, accounting for more than 70 % of total infiltration. Among the three restoration types, Moso bamboo pure forest exhibited the strongest preferential flow, whereas Chinese fir pure forest showed a comparatively higher contribution of matrix flow. Differences in infiltration behavior were closely linked to variations in soil bulk density, non-capillary porosity, texture, and fine root biomass. Preferential flow was primarily governed by non-capillary porosity, fine root biomass (<2 mm), and bulk density. Higher non-capillary porosity and greater fine root biomass promoted the formation of continuous macropore networks and root-induced flow pathways, thereby enhancing preferential flow, while increased bulk density exerted a strong inhibitory effect. In contrast, matrix flow was mainly controlled by fine root biomass, clay content, and bulk density. Higher clay content and bulk density favored water retention and slowed infiltration, thereby increasing the relative importance of matrix flow. Overall, this study highlights that vegetation restoration type is a key determinant of soil infiltration pathways and hydrological functioning. By clarifying the differential controls on preferential flow and matrix flow, the results provide important insights for forest management, hydrological modeling, and soil and water conservation in subtropical regions. Managing vegetation composition and root traits can be an effective strategy to regulate infiltration processes, reduce runoff and erosion, and improve ecosystem water regulation capacity.

Author declarations

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Lin Zhang: Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft. **Tianheng Zhao:** Formal analysis, Writing – original draft. **Shi Qi:** Formal analysis, Funding acquisition, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rhisph.2026.101258>.

Data availability

The data that has been used is confidential.

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